

This section combines our active cloud spending analysis and the migration roadmap to fully exploit the **Microsoft Startups Founders Hub Grant** (representing up to **\$150,000 in Azure Cloud credits** and **\$2,500 in OpenAI credits**) to reduce cash burn to absolute zero.

1. Active Operational Run Rate

Following the conclusion of our trial periods, we have updated our active cloud ledger to reflect commercial operational costs. Our monthly operational burn rate is now **\$1,090.00** for active geospatial imagery, fraud auditing, server compute, and ledger gas.

Infrastructure / Service	Vendor	Current Monthly Cost	Funding Phase Strategy
Intranet & Web Hosting	Vercel (Free Tier)	\$0.00	Upgrade to Vercel Pro (\$20/mo) post-funding for team access.
Database & Scaling	GCP Firestore / Blaze Plan	\$100.00	Scales dynamically; covers production compute and database queries.
Object Storage (Media/PII)	Cloudflare R2	\$0.00	Selected specifically to eliminate AWS/GCP data egress fees.
Geospatial & Convective Mapping	XWeather API & Vexcel Imagery	\$750.00	Commercial subscription for severe convective storm mapping and high-res post-storm imagery.
Core AI Routing & HITL	Google Gemini 1.5 & Pro	\$65.00	Production Gemini API tokens for IoT routing and human-in-the-loop validation.
Claims Fraud Audit	ISO Verisk ClaimSearch	\$150.00	Active double-claim checking API subscription.
Secure Registry & Contracts	Secure L2 Ledger Mainnet	\$25.00	Sponsored gas for transaction settlement.
Version Control	GitHub (Free Tier)	\$0.00	Upgrade to GitHub Team (\$4/user) when engineering team expands.

Infrastructure / Service	Vendor	Current Monthly Cost	Funding Phase Strategy
Total Monthly Burn		\$1,090.00	
2. Microsoft Grant Migration Architecture			
By mapping GCP services to Azure equivalents, we migrate our active workloads to Microsoft's sponsored grant platform:			
GCP Component	Azure Grant Equivalent	Sponsorship / Grant Method	Value & Cash Savings
Vertex AI (Gemini 2.5)	Azure OpenAI Service (GPT-4o & mini)	OpenAI Credit Portal	Replaces Gemini calls with GPT-4o-mini for claims validation.
BigQuery (Telemetry)	Azure Data Lake + Azure Synapse	Azure Grant Credits	Stores signed local IoT weather telemetry history.
Firestore (Metadata & Audits)	Azure Cosmos DB (Mongo API)	Azure Grant Credits	Houses risk scores, FL Bar Numbers, and audit trails.
Cloud Run (Microservices)	Azure Container Apps (ACA) / AKS	Azure Grant Credits	Hosts the Quant Brain Rust sandbox and Vigilance Grid servers.
Cloud PubSub (Ingestion)	Azure Event Hubs	Azure Grant Credits	Receives telemetry payloads from Sentinel weather gateways.
Google Maps / Geospatial	Azure Maps SDK + Esri ArcGIS	Azure Grant Credits	Renders threat boundaries and rooftop structural details.
GitHub Private Seats	GitHub Enterprise	20 Free Seats for 1 Year	Saves ~\$5,000/year in private organization licenses.

3. Step-by-Step Migration Plan

- **Phase 1 (Immediate):** Claim GitHub Enterprise seats; setup branch protections and Actions runners on Azure VMs.
- **Phase 2 (Telemetry Ingest):** Deploy Azure Event Hubs; reconfigure the NodeJS Chainmail-Vigilance-Grid to ingest data on Azure ACA.
- **Phase 3 (Actuarial Engine):** Migrate metadata to serverless Cosmos DB; containerize the Rust Quant Brain; shift LLM calls to Azure OpenAI.
- **Phase 4 (State Reporting):** Setup Azure Functions on timer triggers to automate QUASR/PCLR reports.